

you are a expert statistics teacher.create a details theoretical note about random variable and What are Probability Distributions . seperate module.add a case study in business scenario about the interpretation and usecase also

Imagine standing by the window of a quiet village bakery early in the morning. The baker never knows the exact number of loaves people will buy that day. On some rainy mornings, five loaves sell; on festive mornings, forty disappear before noon. The number changes because of chance, mood, and weather.

In statistics, we do not treat this unknown as complete darkness. Instead, we give this unpredictable number a name, study its habits, and understand how it behaves.

## Module 1: The Random Variable

### The Story: The Baker's Chalkboard

Every morning, the village baker, Lucas, looks at the sky. Many events could happen: "sunny day," "heavy rain," "market day," or "quiet Monday." These are real-world happenings, not numbers.

To run his accounts, Lucas does not write words. He writes down the outcome that matters to his wallet: how many customers walk through the door.

If three people walk in, he writes **3**. If ten walk in, he writes **10**. Lucas built a bridge that takes a messy, real-world happening and turns it into a clear number.

### Modern Parallel

A **Random Variable** (written as an uppercase letter like  $X$ ) is not a fixed number like 5. It is a rule or mapping that assigns a real number to an uncertain real-world outcome.

**The Event (Real World):** A user clicks an online ad, leaves the page, or buys a product.

**The Random Variable ( $X$ ):** The total money spent by that user in dollars.

We divide random variables into two natural families:

**Discrete Random Variables:** Numbers you count in whole steps. You can sell 3 loaves of bread or 4 loaves, but never 3.42 loaves. Modern examples include website visits, defective items in a batch, or customer support calls.

### The Story: The Baker’s Weathered Ledger

After twenty years of baking, Lucas noticed a clear pattern. He rarely sold only 2 loaves, and he almost never sold 100. Most days, sales hovered gently between 25 and 35 loaves.

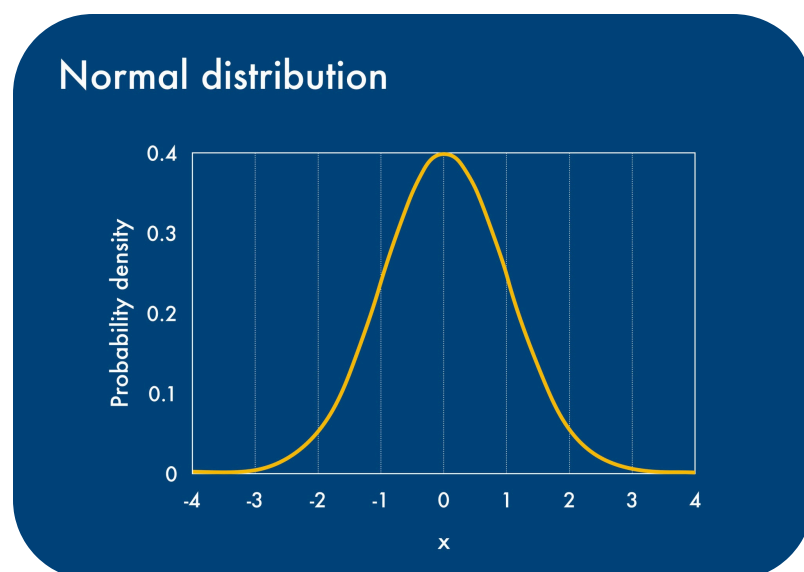
He drew a chart on the cellar wall. Along the floor, he listed every possible number of loaves. Above each number, he piled flat stones for how often that number occurred. The stones formed a hill: thin at the low ends, tall and thick in the center.

That hill of stones is a **Probability Distribution**.

### Modern Parallel

A probability distribution is the complete mathematical "blueprint" of a random variable. It tells you two things:

1. Every possible value the variable can take.
2. How likely each of those values is to happen.



*Normal distribution curve showing common center values. Source: Dimitrios Karamitros / Getty Images*

Notice how the peak in the visual sits right over the center. The probability density is highest where events occur most frequently, while the tails taper off smoothly toward the rare extremes.

## The Two Ways Distributions Behave

**For Discrete Variables (Probability Mass Function):** Probabilities sit in distinct blocks. Each possible outcome gets an exact probability score between 0 and 1, and all scores add up to 100% (1.0).

**For Continuous Variables (Probability Density Function):** Because measurements can take infinite decimal places, the chance of an exact number (like a package arriving at *exactly* 24.000000 minutes) is technically 0. Instead, probability is measured as the **area under the curve** over a range of values (for example, the chance a delivery takes between 20 and 25 minutes).

## Business Case Study: Inventory at FreshBite

FreshBite is a meal-kit startup shipping fresh pre-portioned dinners across a major city. Perishable herbs spoil in 48 hours, so overstocking causes direct food waste. Understocking leads to canceled orders, refunds, and lost customer trust.

### 1. Problem Identification

Daily demand for family meal boxes is unpredictable. Daily orders fluctuate between 180 and 320 boxes. Treating demand as a single fixed average (e.g., "we average 250 boxes, so let's prepare 250") caused FreshBite to stock out on nearly half of all weekdays.

### 2. Implementation with Random Variables

**Random Variable ( $X$ ):** Daily customer orders for family meal boxes.

**Distribution Model:** Historical data shows that daily demand follows a Normal Distribution with an average ( $\mu$ ) of **250 boxes** and a standard deviation ( $\sigma$ ) of **25 boxes**.

### 3. Business Decision & Interpretation

Rather than guessing, management set an operational target: "**We want a 97.5% service level**" (meaning FreshBite satisfies all customer orders 97.5 days out of 100).

Using the properties of the distribution curve:

Around 95% of demand falls within two standard deviations ( $\pm 50$  boxes) of the mean: between 200 and 300 boxes.

Demand exceeds 300 boxes on only about 2.5% of days.

By stocking **300 boxes**, FreshBite protects against stockouts on 97.5% of business days while keeping excess inventory capped at a known, manageable risk.

## Technical Appendix: Math, Code, and Visualization

### Core Mathematical Rules

#### Discrete Random Variable:

For values  $x_1, x_2, \dots$ :

$$\sum_i P(X = x_i) = 1, \quad 0 \leq P(X = x_i) \leq 1$$

Expected value (Mean):

$$E[X] = \mu = \sum_i x_i P(X = x_i)$$

#### Continuous Random Variable:

For a probability density function  $f(x)$ :

$$\int_{-\infty}^{\infty} f(x) dx = 1, \quad f(x) \geq 0$$

Probability within an interval  $[a, b]$ :

$$P(a \leq X \leq b) = \int_a^b f(x) dx$$

Expected value:

$$E[X] = \mu = \int_{-\infty}^{\infty} x f(x) dx$$

### Python Code Syntax & Distribution Plotting

The script below models FreshBite's demand distribution, computes the 97.5% safety threshold, and plots the curve.

Python



```
import numpy as np
import matplotlib.pyplot as plt
from scipy import stats

# 1. Parameter definition
mean_demand = 250      # Average daily orders (mu)
std_demand = 25        # Standard deviation (sigma)
service_level = 0.975  # 97.5% desired fulfillment rate

# 2. Threshold calculation
inventory_target = stats.norm.ppf(service_level, loc=mean_demand, scale=std_demand)
print(f"Target Inventory for 97.5% Service Level: {inventory_target:.1f} boxes")
```

```
# 3. Visualization
x = np.linspace(160, 340, 1000)
y = stats.norm.pdf(x, loc=mean_demand, scale=std_demand)

plt.figure(figsize=(9, 4.5))
plt.plot(x, y, color="#1f77b4", lw=2, label="Daily Demand PDF")

# Shading covered demand (97.5% service area)
x_fill = np.linspace(160, inventory_target, 500)
y_fill = stats.norm.pdf(x_fill, loc=mean_demand, scale=std_demand)
plt.fill_between(x_fill, y_fill, color="#1f77b4", alpha=0.25, label="Covered Demar

# Marker for the inventory policy
plt.axvline(inventory_target, color="#d62728", linestyle="--", lw=1.5,
            label=f"Stock Limit ({inventory_target:.0f} boxes)")

plt.title("FreshBite: Demand Distribution & Inventory Policy", fontsize=12)
plt.xlabel("Daily Box Demand (Units)", fontsize=10)
plt.ylabel("Probability Density", fontsize=10)
plt.legend(frameon=True)
plt.grid(axis='x', linestyle=':', alpha=0.6)
plt.tight_layout()
plt.show()
```

Want to explore how Poisson distributions handle rare events like server crashes?

↔ Yes